

Basic Terms in Graph Theory

Loop: A loop is an edge that connects a vertex to itself.

Multiple Edges: Multiple edges are two or more edges that connect the same pair of vertices.

Endpoints of an edge: The endpoints of an edge are the two vertices that an edge connects.

Adjacency: Two vertices u and v in an undirected graph G are said to be adjacent if there is an edge $e = \{u, v\}$ in G that connects u and v .

Incidence: An edge e is said to be incident with a vertex v if v is one of the endpoints of e .

Degree: The degree of a vertex v , denoted $\deg(v)$, is the number of edges incident with v in an undirected graph.

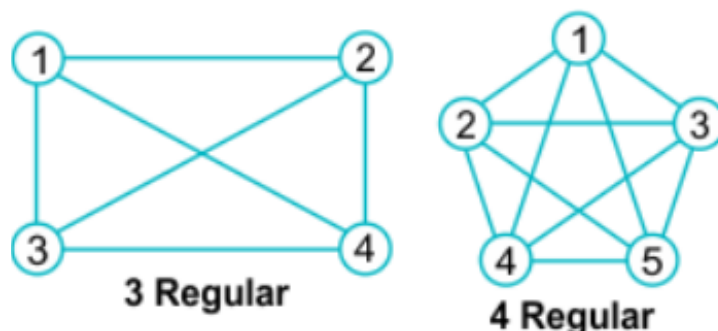
Isolated Vertex: A vertex with degree zero (not connected to any other vertex) is called an isolated vertex.

Pendant Vertex: A vertex with degree one is called a pendant vertex.

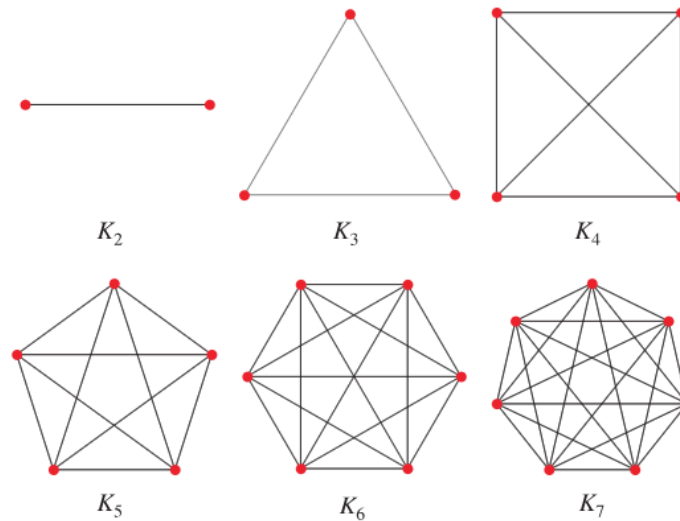
Initial and Terminal Vertex (Directed Graph): In a directed graph, for an edge (u, v) , u is the initial vertex and v is the terminal vertex.

In-degree and Out-degree: In a directed graph, the in-degree of a vertex is the number of edges coming into the vertex, and the out-degree is the number of edges going out of the vertex.

Regular Graph. A *regular graph* is a simple graph in which every vertex has the same degree. If every vertex has degree r , then the graph is called an *r -regular graph*.

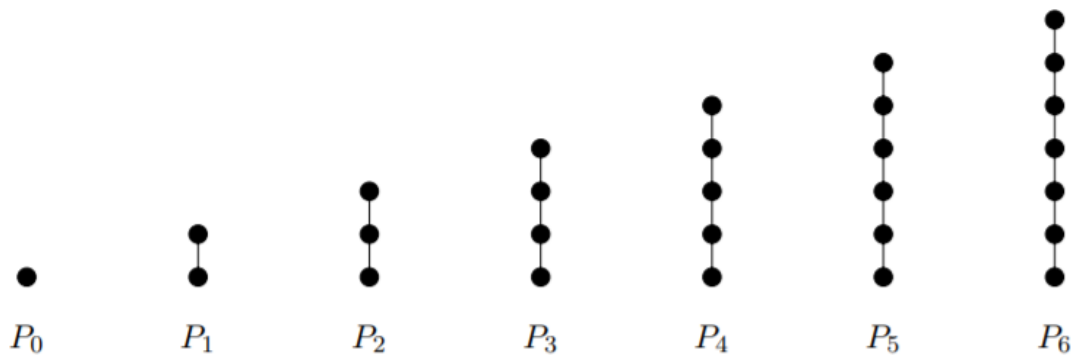


Complete Graph: A complete graph K_n is a simple undirected graph in which every pair of distinct vertices is connected by a unique edge.

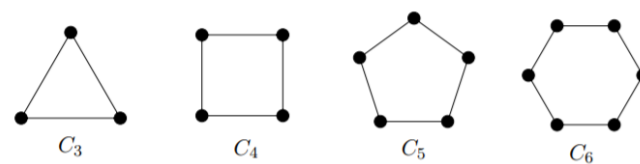


Path: A path P_n is a simple graph consisting of n vertices $v_1, v_2, \dots, v_n$, and $n - 1$ edges of the form

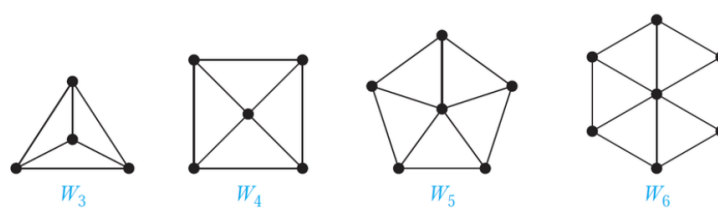
$$\{v_1, v_2\}, \{v_2, v_3\}, \dots, \{v_{n-1}, v_n\}.$$



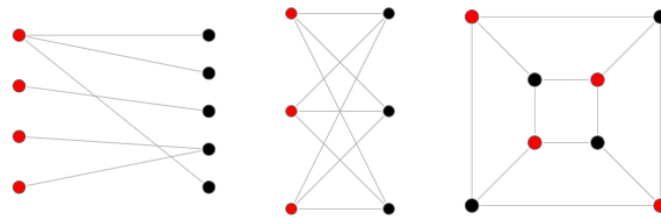
Cycle: A cycle is a closed path in which all edges and all intermediate vertices are distinct.



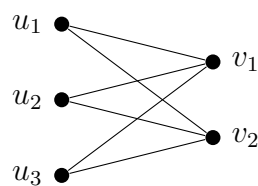
Wheel Graph: A wheel graph W_n is formed by connecting a single vertex to all vertices of a cycle C_n .



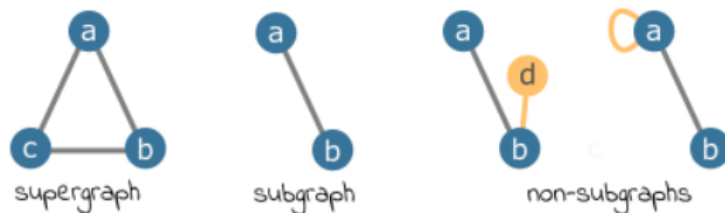
Bipartite Graph: A graph is bipartite if its vertex set can be divided into two disjoint sets V_1 and V_2 such that every edge connects a vertex in V_1 to one in V_2 .



Complete Bipartite Graph: A complete bipartite graph $K_{m,n}$ is a bipartite graph where every vertex in V_1 is connected to every vertex in V_2 .

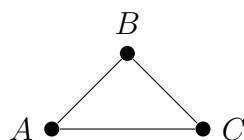


Subgraph: A subgraph of a graph $G = (V, E)$ is a graph $H = (V', E')$ such that $V' \subseteq V$ and $E' \subseteq E$.



Exercise

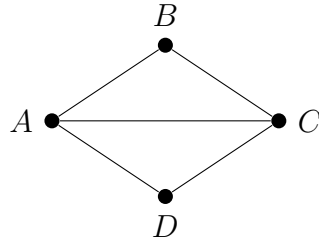
Question: Is the following graph bipartite?



Answer: No, it is not bipartite. A cycle of length 3 (odd length) cannot be bipartite because we cannot split the vertices into two disjoint sets without connecting vertices within the same set.

Handshaking Lemma

Statement: In any undirected graph, the sum of the degrees of all vertices is equal to twice the number of edges.



Example:

Vertex degrees: $A = 2$, $B = 2$, $C = 2$, $D = 2$. Total = 8.

Number of edges = 4.

Hence, $2|E| = \sum \deg(v) = 8$.

Proof: Each edge in an undirected graph contributes exactly 2 to the total degree count (one for each endpoint), so the sum of degrees equals $2|E|$.

Theorem: Even Number of Odd Degree Vertices

Statement: In any undirected graph, the number of vertices with odd degree is even.

Proof: From the handshaking lemma, we know the total sum of all vertex degrees is even. Suppose there were an odd number of vertices with odd degree; their sum would be odd, which contradicts the total being even. Hence, the number of odd-degree vertices must be even.